

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – 26085
- Problem Statement Title- Urban Flood Nowcasting System (Drainage and Rainfall Coupling)
- Theme- Disaster Management
- PS Category- Software
- Team ID- 144020
- Team Name - Flood Busters





Flood Busters

NALANETRA FLOODGRID



SMART INDIA
HACKATHON
2026

PROBLEM OVERVIEW & GROUND REALITY

- **Economic Impact:** Urban flooding damages property and disrupts city services.
- **Critical Delays:** Flooded roads and underpasses delay commuters and emergency access.
- **Blind Forecasts:** Street-level risk needs rainfall, terrain and local drainage data.
- **Ghost Closures:** Closure claims need field evidence and officer verification.
- **Emergency Hazard:** Flooded corridors can obstruct commuters and ambulances.

PROPOSED SOLUTION (NALANETRA ECOSYSTEM)

- **Predictive Layer (Doppler + DEM):** Target: 3-hr hyper-local flood-risk nowcast.
- **Ground Truth Validation:** 1-Tap citizen reporting with GPS & depth tags.
- **Dynamic Conflict Resolution:** Officer-reviewed visual evidence updates incident priority.
- **Proof-Gated Closure Loop:** Field crews submit before/after photo proof.
- **Graph Drainage Hydraulics:** Manholes (nodes) & stormwater pipes (edges) modeled.

GURUGRAM: 9-10 JULY 2025



133 mm

Rainfall in 12 hours

103 mm

Of this in 90 minutes

Waterlogged streets and disrupted access

[Rainfall: DDMA advisory, reported by PTI \(10 July 2025\)](#)

[Photo: PTI / Mint. Gurugram, July 2025](#)

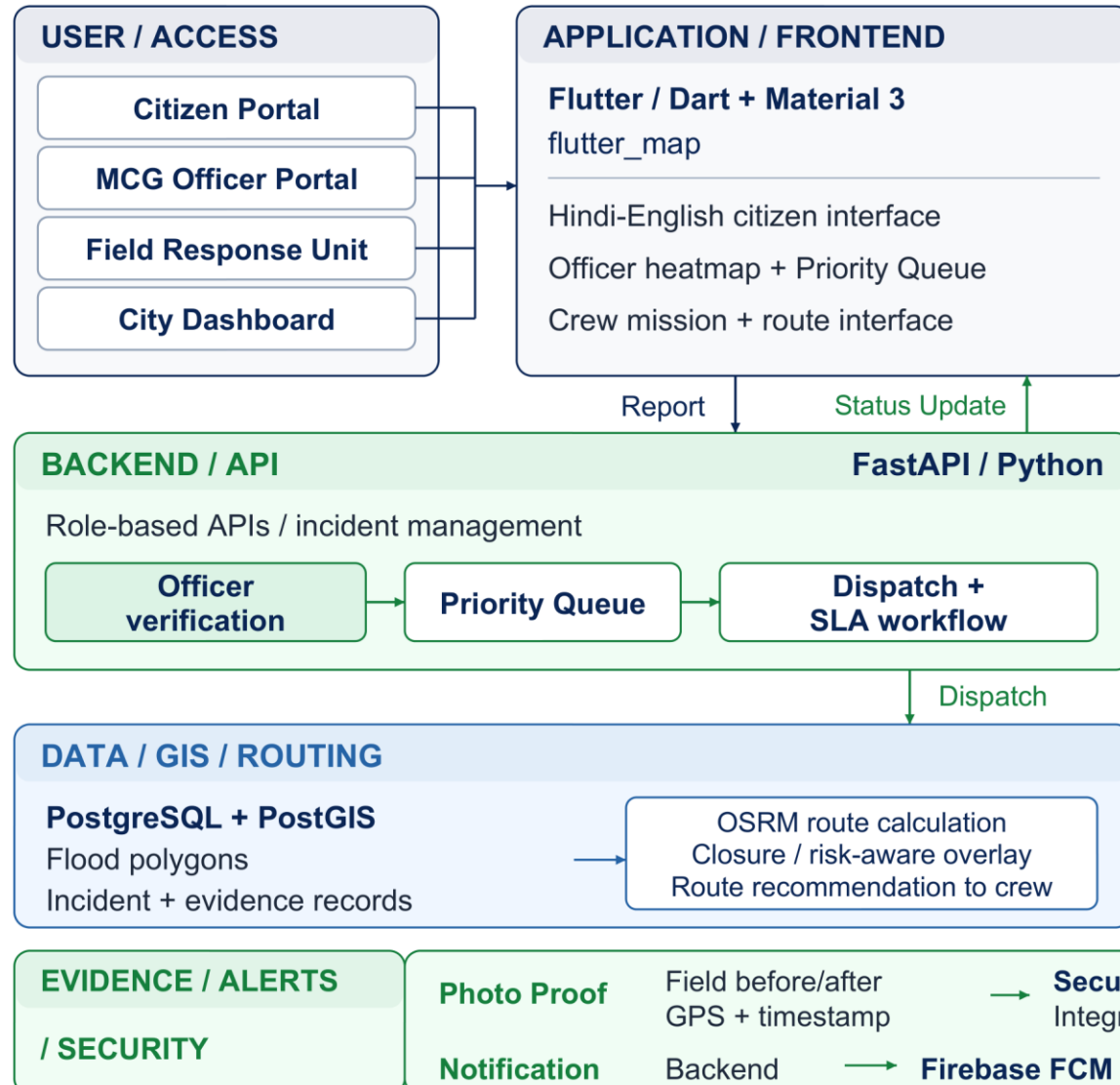


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TECHNICAL APPROACH



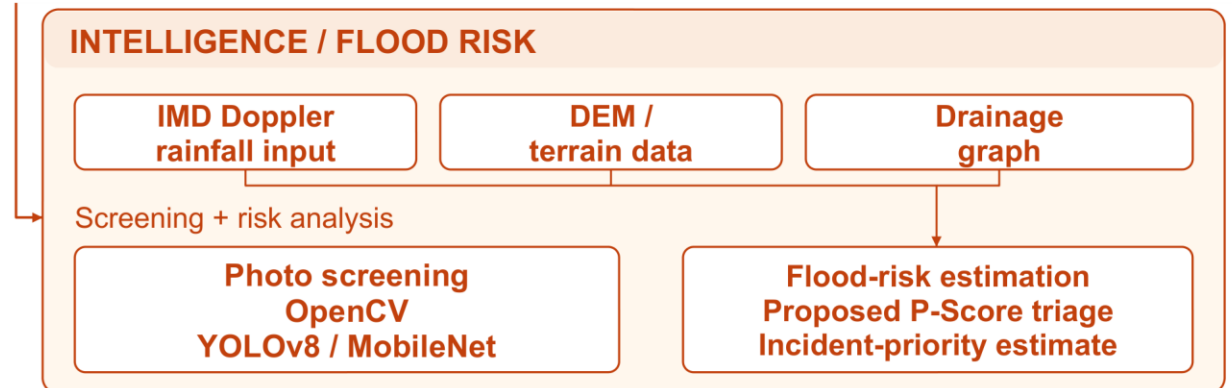
NALANETRA SYSTEM ARCHITECTURE



SCIENTIFIC PRIORITY FORMULA (MATHEMATICAL MODEL)

$$P = 0.30(S) + 0.20(R) + 0.15(W) + 0.15(D) + 0.10(E) + 0.10(A)$$

- **S — Severity Depth (30%)**: Verified water-depth severity.
- **R — Rainfall Severity (20%)**: Radar-based rainfall intensity.
- **W — Ward Criticality (15%)**: Population and civic-asset exposure.
- **D — Drainage Deficit (15%)**: Capacity loss from blockage or siltation.
- **E — Emergency Importance (10%)**: Hospital-route importance.
- **A — SLA Aging Factor (10%)**: Pending time relative to the SLA.
- Inputs use 0-1. Higher means greater urgency. Weights need calibration.
- **P ranks dispatch priority**. It does not predict flood depth.
- **Coupling** uses radar rainfall + DEM runoff + drainage capacity to estimate flood risk.





FEASIBILITY AND VIABILITY



Flood Busters

Feasibility:

1. Technical Feasibility:

- Doppler, DEM and drainage modeling form the proposed stack.
- Open-source stack (Python, PostGIS, OSRM) — no vendor lock-in.

2. Economical Feasibility:

- Zero sensor capex — uses existing IMD Doppler & DEM data feeds.
- Pilot costs need cloud, map/API and support estimates.

3. Social Feasibility:

- Bilingual Hindi/English 1-tap app for any smartphone user.
- Community-driven reporting empowers active citizen participation.

4. Legal & Compliance:

- Design informed by NDMA Urban Flooding Guidelines, 2010.
- Consent and PII safeguards need review before deployment.

Viability:

1. Operational Viability:

- AI triage supports officer review and crew dispatch.
- Municipal GIS/SCADA integration needs data-access agreements.

2. Security & Audit Trail:

- SHA-256 hashes help detect changes to recorded evidence.
- Role-based access control (RBAC) across all three portals.

3. Scalability & Replication:

- City rollout needs local drainage data and model calibration.
- Planned open APIs can support municipal integration.

4. Financial Sustainability:

- Evaluate municipal contracts and PPP funding options.
- Reuse city infrastructure, with hosting and operating costs.

COMPETITIVE EDGE MATRIX

Key Capability	NalaNetra (Ours)	Municipal 1800	Google FloodHub
Drainage-Rain Coupling	✓ Dynamic Coupling	✗ Manual	✗ Basin Only
Hyper-Local Street Truth	✓ 1-Tap GPS Truth	✗ Voice Calls Only	✗ None
Proof-Gated Photo Closure	✓ Mandatory Proof	✗ High Ghost Risk	✗ None
Flood-Aware Safe Routing	✓ Flood-Aware Routing	✗ No Routing	▲ Partial

POTENTIAL CHALLENGES & MITIGATION STRATEGIES

- **Monsoon Connectivity Blackout:** Offline storage queues photos; auto-syncs online. Offline caching & store-and-forward mechanism.
- **Fake / Spam Citizen Reports:** AI photo-relevance screening with mandatory officer verification.
- **Duplicate Report Overhead:** Auto-clusters reports within 150m and 6 hours into one incident.
- **Weather API Rate Limits:** Local caching of Doppler rasters with resilient heuristic fallback.



IMPACT AND BENEFITS



Flood Busters

Impact:

1. Rapid Response — TARGET: 80% FASTER

- Target lower median dispatch-to-arrival time in the pilot.
- Compare verified response times with a pilot baseline.

2. Proof-Linked Closures

- GPS-tagged before/after photo evidence for every closure.
- Officer review helps detect incomplete closure evidence.

3. Optimized Logistics — TARGET: -70% Trips

- 150m / 6-hour grouping reduces redundant crew trips.
- Validate grouping to avoid merging distinct flood incidents.

4. Early Warning — TARGET: 3 HOURS

- Couple Doppler rainfall, DEM runoff and drainage capacity.
- Evaluate forecast skill up to a three-hour lead time.
- Impact figures are pilot targets, pending field validation.

Benefits:

1. Enhanced Public Safety

- Protects commuters from underpass flooding.
- Flood alerts support decisions using current observations.

2. Hospital Route Protection

- Flag flooded roads along hospital and ambulance corridors.
- Flood-aware OSRM routing for emergency vehicles.

3. Municipal Accountability

- Evidence-linked workflow from complaint to closure.
- SLA dashboards track officer & crew performance.

4. Citizen Trust & Transparency

- Live five-stage tracking improves status visibility.
- Citizens see real-time status from report to resolution.

5. Resilient Urban Planning

- Historical flood analytics guide AMRUT 2.0 drainage.
- Data-driven investment for flood-resilient cities.

NATIONWIDE SCALING ROADMAP

PHASE 1: PILOT WARD

Months 1–3 • Proposed Ward 14 pilot

- Confirm radar and drainage-data access.
- Test coupled risk and citizen reporting.
- Measure forecast skill and response time.



PHASE 2: CITY-WIDE EXPANSION

Months 4–8 • City-wide expansion after pilot validation

- Expand ward by ward after calibration.
- Test flood-aware OSRM route updates.
- Seek Traffic Police data agreements.
- Track adoption, SLA and operating costs.



PHASE 3: NATIONAL SCALING

Months 9–18 • Replication to other cities after validation

- Onboard local terrain and drainage data.
- Validate forecasts separately in each city.
- Provide open APIs and staff training.
- Confirm funding and operating partners.

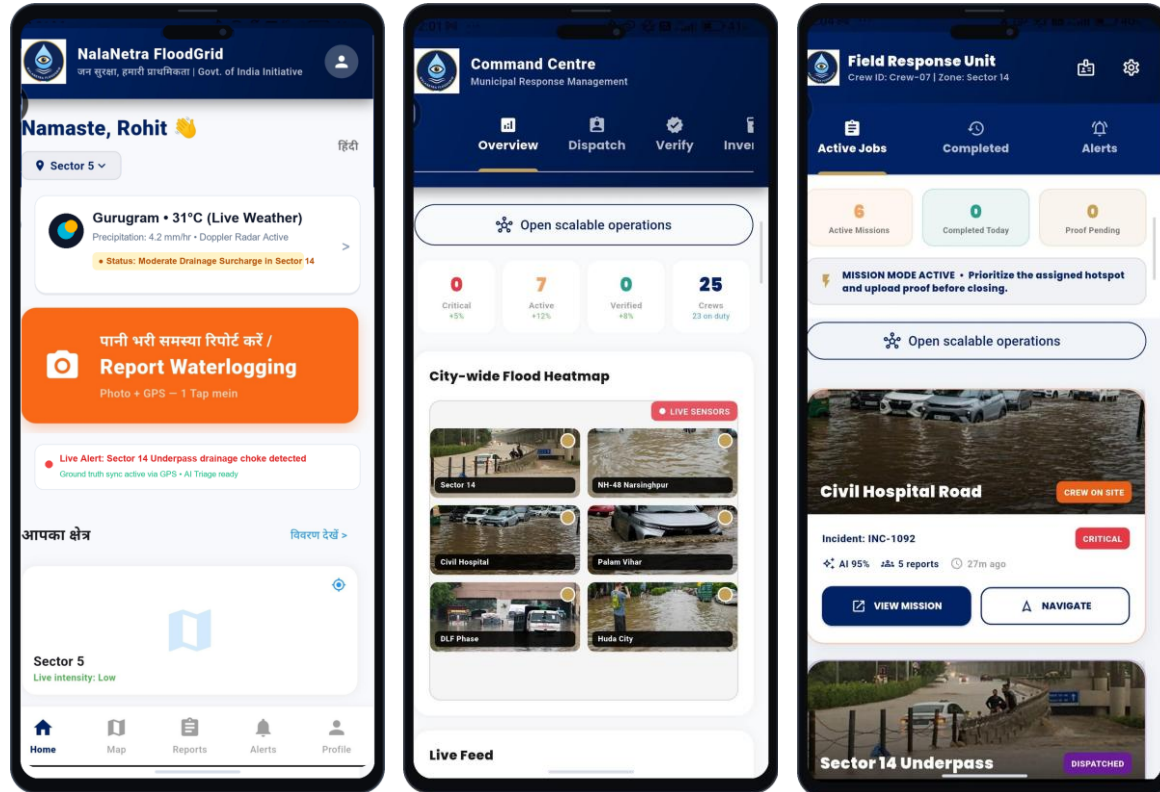


RESEARCH & REFERENCES



Flood Busters

OUR WORKING PROTOTYPE [NalaNetra GitHub Repo](#)



Citizen Portal (Hindi UI)

1-Tap GPS Report,
Live Weather
& 5-Stage Stepper

MCG Officer Portal

City Heatmap, P-Score
Priority
Queue & Crew
Dispatch

Field Response Unit

Turn-by-Turn Safe
Navigation
& Mandatory Photo
Proof

SCIENTIFIC RESEARCH & TECHNICAL FOUNDATIONS (MoES & INDIA)

- 1. IMD Nowcast Services:** Short-lead weather forecasts inform the proposed warning horizon. [View IMD Guide](#)
- 2. IMD Radar Services:** Radar observations inform rainfall inputs. Data access and local coverage need confirmation. [View IMD Radar](#)
- 3. EPA SWMM Drainage Modeling:** Runoff and pipe-network methods inform rainfall-drainage coupling. [View EPA SWMM](#)
- 4. OSRM Routing Documentation:** Edge-weight updates inform flood-aware route design. [View OSRM Docs](#)

STATUTORY GOVERNANCE, POLICY & TECH STANDARDS (GOVT. OF INDIA)

- **NDMA Urban Flooding Guidelines:** Drainage capacity, maintenance and warning context for our problem. [View NDMA](#)
- **DPDP Act, 2023 (MeitY):** Privacy reference for citizen data. Deployment needs a compliance review. [View DPDP Act](#)
- **NIST Hash Functions:** SHA-256 supports evidence integrity checks. Hashing does not encrypt data. [View NIST](#)
- **OWASP MASVS Security Standard:** Storage and access-control tests guide security validation. [View OWASP](#)